

DERIVATION OF DC-LINK CAPACITOR AND DOUBLE-FREQUENCY RIPPLE POWER

The DC-bus voltage is represented by a steady-state component and a second-harmonic (double-frequency) ripple component as

$$V_g^{\text{DC}}(t) = V_g^{\text{DC},0} + \underbrace{V_g^{\text{DC},2\omega} \sin(2\omega t + \theta_{v,g}^{\text{DC}})}_{\text{double-frequency voltage ripple}}. \quad (1)$$

Here, $V_g^{\text{DC}}(t)$ is the instantaneous DC-bus voltage of converter g , $V_g^{\text{DC},0}$ is its steady-state (average) component, $V_g^{\text{DC},2\omega}$ is the amplitude of the double-frequency voltage ripple, ω is the fundamental grid angular frequency, and $\theta_{v,g}^{\text{DC}}$ is the phase angle of the double-frequency voltage ripple.

Similarly, the DC-bus current is represented as

$$I_g^{\text{DC}}(t) = I_g^{\text{DC},0} + \underbrace{I_g^{\text{DC},2\omega} \sin(2\omega t + \theta_{i,g}^{\text{DC}})}_{\text{double-frequency current ripple}}, \quad (2)$$

where $I_g^{\text{DC}}(t)$ is the instantaneous DC-bus current, $I_g^{\text{DC},0}$ is its average component, $I_g^{\text{DC},2\omega}$ is the amplitude of its double-frequency component, and $\theta_{i,g}^{\text{DC}}$ is the corresponding current-ripple phase angle.

The voltage- and current-ripple phase angles are denoted separately by $\theta_{v,g}^{\text{DC}}$ and $\theta_{i,g}^{\text{DC}}$ to account for a possible phase displacement between the double-frequency voltage and current components.

Double-Frequency DC-Bus Power

The instantaneous DC-bus power is obtained from the product of the instantaneous DC-bus voltage and current:

$$\begin{aligned} P_g^{\text{DC}}(t) &= V_g^{\text{DC}}(t) I_g^{\text{DC}}(t) \\ &= [V_g^{\text{DC},0} + V_g^{\text{DC},2\omega} \sin(2\omega t + \theta_{v,g}^{\text{DC}})] \\ &\quad \times [I_g^{\text{DC},0} + I_g^{\text{DC},2\omega} \sin(2\omega t + \theta_{i,g}^{\text{DC}})] \\ &= \underbrace{V_g^{\text{DC},0} I_g^{\text{DC},0}}_{\text{average-power contribution}} \\ &\quad + \underbrace{V_g^{\text{DC},0} I_g^{\text{DC},2\omega} \sin(2\omega t + \theta_{i,g}^{\text{DC}})}_{\text{double-frequency contribution}} \\ &\quad + \underbrace{V_g^{\text{DC},2\omega} I_g^{\text{DC},0} \sin(2\omega t + \theta_{v,g}^{\text{DC}})}_{\text{double-frequency contribution}} \\ &\quad + \underbrace{V_g^{\text{DC},2\omega} I_g^{\text{DC},2\omega} \sin(2\omega t + \theta_{v,g}^{\text{DC}}) \sin(2\omega t + \theta_{i,g}^{\text{DC}})}_{\text{ripple-ripple coupling contribution}} \end{aligned} \quad (3)$$

The product of the two double-frequency ripple components in the last term of (3) produces a DC component and a 4ω component. Hence, it does not contribute directly to the 2ω component of the DC-bus power.

Therefore, isolating only the terms oscillating at 2ω gives

$$\begin{aligned} P_g^{\text{DC},2\omega}(t) &= \underbrace{V_g^{\text{DC},0} I_g^{\text{DC},2\omega} \sin(2\omega t + \theta_{i,g}^{\text{DC}})}_{\text{average voltage} \times \text{current ripple}} \\ &\quad + \underbrace{V_g^{\text{DC},2\omega} I_g^{\text{DC},0} \sin(2\omega t + \theta_{v,g}^{\text{DC}})}_{\text{voltage ripple} \times \text{average current}}. \end{aligned} \quad (4)$$

Equation (4) represents the double-frequency power oscillation at the DC bus. The amplitude of this oscillatory component is hereafter denoted by $P_g^{2\omega}$.

Relationship Between DC-Bus Ripple Power and Capacitor Ripple Power

The double-frequency power oscillation at the DC bus results in a periodic energy imbalance that must be buffered by the DC-link energy-storage element. In the present derivation, the DC-link is assumed ideal and lossless, and the DC-link capacitor is assumed to be the element that buffers the dominant double-frequency power oscillation.

Accordingly, the instantaneous capacitor power may have a sign opposite to the DC-bus power imbalance depending on the adopted power-flow convention, since the capacitor alternately absorbs and releases energy. However, under the ideal DC-link assumption, the magnitudes of their double-frequency power amplitudes are equal. Therefore,

$$P_g^{2\omega} = P_{C,g}^{2\omega}, \quad (5)$$

where $P_g^{2\omega}$ is the amplitude of the double-frequency DC-bus power identified from (4), and $P_{C,g}^{2\omega}$ is the corresponding double-frequency power amplitude buffered by the DC-link capacitor.

Equation (5) provides the link between the double-frequency DC-bus power and the capacitor-sizing relationship derived below.

DC-Link Capacitor Current

To relate the double-frequency power to the required DC-link capacitance, the capacitor current is considered separately from the total DC-bus current defined in (2). The DC-link capacitor current is denoted by $I_{C,g}^{\text{DC}}(t)$ and is given by

$$I_{C,g}^{\text{DC}}(t) = C_g \frac{dV_g^{\text{DC}}(t)}{dt}. \quad (6)$$

Substituting (1) into (6) gives

$$\begin{aligned} I_{C,g}^{\text{DC}}(t) &= C_g \frac{d}{dt} [V_g^{\text{DC},0} + V_g^{\text{DC},2\omega} \sin(2\omega t + \theta_{v,g}^{\text{DC}})] \\ &= 2\omega C_g V_g^{\text{DC},2\omega} \cos(2\omega t + \theta_{v,g}^{\text{DC}}). \end{aligned} \quad (7)$$

Hence, the amplitude of the double-frequency capacitor current is

$$I_{C,g}^{\text{DC},2\omega} = 2\omega C_g V_g^{\text{DC},2\omega}. \quad (8)$$

It is important to distinguish the two current quantities used in the derivation. The quantity $I_g^{\text{DC}}(t)$ in (2) denotes the total DC-bus current and is used to determine the DC-bus power, whereas $I_{C,g}^{\text{DC}}(t)$ in (6) denotes specifically the current through the DC-link capacitor and is used to establish the capacitor-sizing relationship.

Double-Frequency Power Buffered by the DC-Link Capacitor

The instantaneous power exchanged with the DC-link capacitor is

$$P_{C,g}^{\text{DC}}(t) = V_g^{\text{DC}}(t) I_{C,g}^{\text{DC}}(t). \quad (9)$$

Substituting (1) and (7) into (9) gives

$$\begin{aligned} P_{C,g}^{\text{DC}}(t) &= [V_g^{\text{DC},0} + V_g^{\text{DC},2\omega} \sin(2\omega t + \theta_{v,g}^{\text{DC}})] \\ &\quad \times [2\omega C_g V_g^{\text{DC},2\omega} \cos(2\omega t + \theta_{v,g}^{\text{DC}})]. \end{aligned} \quad (10)$$

Expanding gives

$$P_{C,g}^{\text{DC}}(t) = \underbrace{2\omega C_g V_g^{\text{DC},0} V_g^{\text{DC},2\omega} \cos(2\omega t + \theta_{v,g}^{\text{DC}})}_{\text{double-frequency capacitor power}} + \underbrace{2\omega C_g (V_g^{\text{DC},2\omega})^2 \sin(2\omega t + \theta_{v,g}^{\text{DC}}) \cos(2\omega t + \theta_{v,g}^{\text{DC}})}_{\text{higher-order ripple contribution}}$$

Using the trigonometric identity

$$\sin(\alpha) \cos(\alpha) = \frac{1}{2} \sin(2\alpha), \quad (12)$$

(11) becomes

$$P_{C,g}^{\text{DC}}(t) = \underbrace{2\omega C_g V_g^{\text{DC},0} V_g^{\text{DC},2\omega} \cos(2\omega t + \theta_{v,g}^{\text{DC}})}_{\text{dominant } 2\omega \text{ component}} + \underbrace{\omega C_g (V_g^{\text{DC},2\omega})^2 \sin(4\omega t + 2\theta_{v,g}^{\text{DC}})}_{\text{higher-order } 4\omega \text{ component}}. \quad (13)$$

Since

$$V_g^{\text{DC},2\omega} \ll V_g^{\text{DC},0}, \quad (14)$$

the second term in (13) is of order $\mathcal{O}((V_g^{\text{DC},2\omega})^2)$ and is neglected under the small-ripple approximation. Therefore, the dominant double-frequency capacitor power is

$$P_{C,g}^{\text{DC},2\omega}(t) \approx 2\omega C_g V_g^{\text{DC},0} V_g^{\text{DC},2\omega} \cos(2\omega t + \theta_{v,g}^{\text{DC}}). \quad (15)$$

Consequently, the amplitude of the double-frequency power buffered by the DC-link capacitor is

$$P_{C,g}^{2\omega} = V_g^{\text{DC},0} I_{C,g}^{\text{DC},2\omega} = 2\omega C_g V_g^{\text{DC},0} V_g^{\text{DC},2\omega}. \quad (16)$$

Finally, from the ideal DC-link ripple-power balance in (5),

$$\begin{aligned} P_g^{2\omega} &= P_{C,g}^{2\omega} \\ &= V_g^{\text{DC},0} I_{C,g}^{\text{DC},2\omega} \\ &= 2\omega C_g V_g^{\text{DC},0} V_g^{\text{DC},2\omega}. \end{aligned} \quad (17)$$

Thus, (17) relates the amplitude of the double-frequency DC-bus power to the DC-link capacitance and the corresponding double-frequency DC-bus voltage-ripple amplitude under the stated ideal DC-link and small-ripple assumptions.

DC-LINK CAPACITOR AND RIPPLE POWER COST MODEL

The DC-link capacitor cost follows the analytical expression in [1], where the cost of a metallized polypropylene film capacitor is expressed as a function of the rated voltage V_0 and capacitance C_g :

$$C_{\text{dc}} = a_{\text{film}} + b_{\text{film}} V_0 + c_{\text{film}} C_g, \quad (18)$$

with $a_{\text{film}} = -1.19574$ \$, $b_{\text{film}} = 0.06429852$ \$/V, and $c_{\text{film}} = 0.00283842$ \$/μF.

Using $V_0 = 1100$ V and $C_g = 40$ μF, the reference capacitor cost is

$$C_{\text{dc}} \approx \$69.72. \quad (19)$$

The reference average cost per unit capacitance at this design point is therefore

$$\begin{aligned} C_{c_g} &= \frac{C_{\text{dc}}}{C_g} \\ &= \frac{69.72}{40} \\ &\approx 1.743 \text{ } \$/\mu\text{F} \\ &\approx 1.743 \times 10^6 \text{ } \$/\text{F}. \end{aligned} \quad (20)$$

For the sizing optimization, this reference average unit capacitance cost is used to construct the linear capacitor-cost approximation

$$\hat{C}_{\text{dc}}(C_g) = C_{c_g} C_g. \quad (21)$$

It should be noted that C_{c_g} is obtained as the reference average unit capacitance cost from (20), rather than as the derivative of the original analytical cost model in (18). However, once the linear approximation in (35) is adopted, its derivative with respect to C_g is

$$\frac{d\hat{C}_{\text{dc}}}{dC_g} = C_{c_g}. \quad (22)$$

Thus, within the adopted linear cost approximation, $d\hat{C}_{\text{dc}}/dC_g$ represents the capacitor cost per unit increase in capacitance and has units of \$/F.

A. Relationship between Capacitance and Ripple Power

From (17), the required DC-link capacitance is

$$C_g = \frac{P_g^{2\omega}}{2\omega V_g^{\text{DC},0} V_g^{\text{DC},2\omega}}. \quad (23)$$

The steady-state and double-frequency components of the DC-bus voltage can be related through the allowable peak-to-peak voltage ripple. From (1), the maximum and minimum DC-bus voltages are

$$V_g^{\text{DC},\text{max}} = V_g^{\text{DC},0} + V_g^{\text{DC},2\omega}, \quad (24)$$

$$V_g^{\text{DC},\text{min}} = V_g^{\text{DC},0} - V_g^{\text{DC},2\omega}. \quad (25)$$

Therefore, the peak-to-peak DC-bus voltage ripple is

$$\begin{aligned} \Delta V_g^{\text{DC},pp} &= V_g^{\text{DC},\text{max}} - V_g^{\text{DC},\text{min}} \\ &= (V_g^{\text{DC},0} + V_g^{\text{DC},2\omega}) - (V_g^{\text{DC},0} - V_g^{\text{DC},2\omega}) \\ &= 2V_g^{\text{DC},2\omega}. \end{aligned} \quad (26)$$

According to [2] and related literature, the maximum allowable peak-to-peak DC-bus voltage ripple is constrained as

$$\Delta V_g^{\text{DC},pp} \leq 0.10 V_g^{\text{DC},0}. \quad (27)$$

For the capacitor-sizing calculation, the ripple constraint is assumed to be active at its maximum allowable value. Therefore,

$$\Delta V_g^{\text{DC},pp} = 0.10 V_g^{\text{DC},0}. \quad (28)$$

Combining (26) and (28) gives

$$2V_g^{\text{DC},2\omega} = 0.10 V_g^{\text{DC},0}, \quad (29)$$

and hence

$$V_g^{\text{DC},2\omega} = 0.05 V_g^{\text{DC},0}. \quad (30)$$

Substituting (30) into (23) gives

$$C_g = \frac{P_g^{2\omega}}{0.1\omega \left(V_g^{\text{DC},0}\right)^2}. \quad (31)$$

Equation (31) therefore expresses the required DC-link capacitance as a linear function of the double-frequency ripple power for the adopted DC-bus voltage-ripple limit.

B. Ripple Power Cost Model

The DC-link ripple-power cost is obtained by recognizing that the adopted capacitor cost is a function of the required capacitance, while the required capacitance is itself a function of the double-frequency ripple power. The functional dependence is therefore

$$\hat{C}_{\text{dc}} = \hat{C}_{\text{dc}}(C_g(P_g^{2\omega})). \quad (32)$$

From (31), the relationship between C_g and $P_g^{2\omega}$ is linear. Therefore, differentiating with respect to $P_g^{2\omega}$ gives

$$\frac{dC_g}{dP_g^{2\omega}} = \frac{1}{0.1\omega \left(V_g^{\text{DC},0}\right)^2}. \quad (33)$$

The quantity $dC_g/dP_g^{2\omega}$ has units of F/W and represents the required increase in DC-link capacitance per unit increase in double-frequency ripple power under the adopted voltage-ripple limit.

To obtain a constant cost coefficient suitable for the optimization, the capacitor cost evaluated at the reference design is expressed as an average cost per unit capacitance,

$$C_{c_g} = \frac{C_{\text{dc,ref}}}{C_{g,\text{ref}}}. \quad (34)$$

Using this coefficient, the DC-link capacitor cost is approximated by the linear relationship

$$\hat{C}_{\text{dc}} = C_{c_g} C_g, \quad (35)$$

where $\hat{C}_{\text{dc}}$ denotes the adopted linear approximation of the capacitor cost. Accordingly,

$$\frac{d\hat{C}_{\text{dc}}}{dC_g} = C_{c_g}. \quad (36)$$

Since $\hat{C}_{\text{dc}}$ is a function of C_g , and C_g is in turn a function of $P_g^{2\omega}$, the DC-link cost per unit of double-frequency ripple power follows from the chain rule:

$$\begin{aligned} \frac{d\hat{C}_{\text{dc}}}{dP_g^{2\omega}} &= \frac{d\hat{C}_{\text{dc}}}{dC_g} \frac{dC_g}{dP_g^{2\omega}} \\ &= \underbrace{C_{c_g}}_{\text{cost per unit capacitance}} \underbrace{\frac{1}{0.1\omega \left(V_g^{\text{DC},0}\right)^2}}_{\text{capacitance per unit ripple power}} \\ &= \frac{C_{c_g}}{0.1\omega \left(V_g^{\text{DC},0}\right)^2}. \end{aligned} \quad (37)$$

Accordingly, the DC-link ripple-power cost coefficient is defined as

$$c_{2\omega} = \frac{d\hat{C}_{\text{dc}}}{dP_g^{2\omega}} = \frac{C_{c_g}}{0.1\omega \left(V_g^{\text{DC},0}\right)^2}. \quad (38)$$

The dimensional consistency of the chain-rule relationship is

$$\underbrace{\left[\frac{d\hat{C}_{\text{dc}}}{dC_g}\right]}_{\$/\text{F}} \underbrace{\left[\frac{dC_g}{dP_g^{2\omega}}\right]}_{\text{F/W}} = \underbrace{\left[\frac{d\hat{C}_{\text{dc}}}{dP_g^{2\omega}}\right]}_{\$/\text{W}}. \quad (39)$$

Using

$$\omega = 2\pi f, \quad f = 60 \text{ Hz}, \quad V_g^{\text{DC},0} = V_0 = 1100 \text{ V}, \quad (40)$$

and

$$C_{c_g} = 1.743 \times 10^6 \text{ \$/F}, \quad (41)$$

the ripple-power cost coefficient becomes

$$\begin{aligned} c_{2\omega} &= \frac{1.743 \times 10^6}{0.1(2\pi \times 60)(1100)^2} \\ &\approx 0.0382 \text{ \$/W} \\ &= 38.2 \text{ \$/kW}. \end{aligned} \quad (42)$$

Thus, $c_{2\omega} = 38.2 \text{ \$/kW}$ represents the adopted linearized DC-link capacitor cost per unit of double-frequency ripple power. It is obtained by combining the adopted average cost per unit capacitance with the dependence of the required DC-link capacitance on the double-frequency ripple power.

REFERENCES

- [1] R. Burkart and J. W. Kolar, "Component cost models for multi-objective optimizations of switched-mode power converters," in *2013 IEEE Energy Conversion Congress and Exposition*, pp. 2139–2146, 2013.
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